

CS 486/686

Unsupervised Learning

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Lecture 17

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Learning goals

- Understand what **unsupervised learning** is.
- Trace the **k-means** clustering algorithm.
- Perform **PCA**: mean-center, covariance, eigendecomposition.
- Understand the basic idea of **autoencoders** and **GANs**.

Two big unsupervised tasks

No labels — only the raw data $\{x_i\}$. What can we learn?

Representation learning



Find a low-dimensional summary of each example.

Clustering, PCA, autoencoders.

Generative modelling



Learn the data distribution so we can *sample new examples*.

GANs, VAEs, diffusion.

Today: one task from each, plus two neural variants.

Clustering: hard vs soft

Group training examples into **clusters**, treating them like discovered categories.

Hard clustering

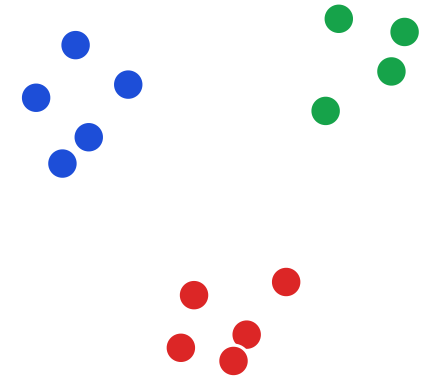
Each x gets
exactly one cluster.

$$\text{class}(x) = c$$

Soft clustering

Each x gets a
distribution over
clusters.

$$\text{class}(x) \sim P(C \mid x)$$



Each color is a discovered cluster — no labels were given.

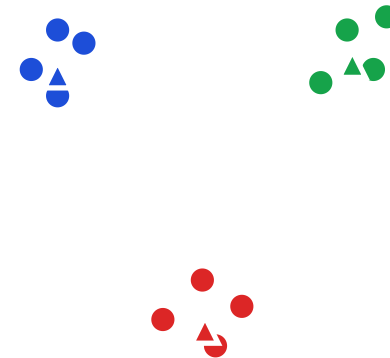
k-Means: centroids and distance

Inputs: number of clusters k and m training examples $x_i \in \mathbb{R}^n$. Each cluster has a **centroid** $c \in \mathbb{R}^n$; each x_i is assigned to its closest centroid.

Euclidean distance (L2)

$$d(c, x) = \sqrt{\sum_{j=1}^n (c_j - x_j)^2}$$

- **Hard clustering:** each example goes to one centroid.
- Centroids live in the same feature space as the data.



Triangles = current centroids; circles = data points.

The k-means algorithm

Alternate two steps until convergence:

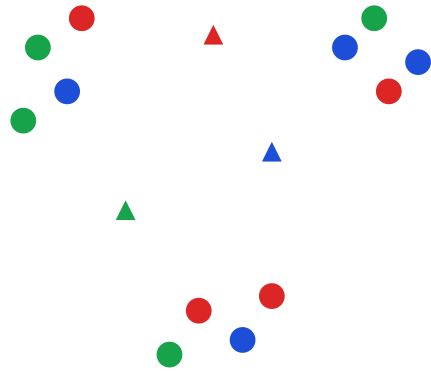
$\text{k-means}(X, k)$

1. Initialize k centroids $C[0 \dots k - 1]$ randomly.
2. Repeat until assignments stop changing:
 - **Assign step:** for each example i , $y[i] \leftarrow \arg \min_c d(C[c], X[i])$.
 - **Update step:** for each cluster c , $C[c] \leftarrow \text{mean of all } X[i] \text{ with } y[i] = c$.
- Guaranteed to converge with L2 distance.
- Not guaranteed to find the global optimum.
- **Mitigate:** random restarts, feature scaling.

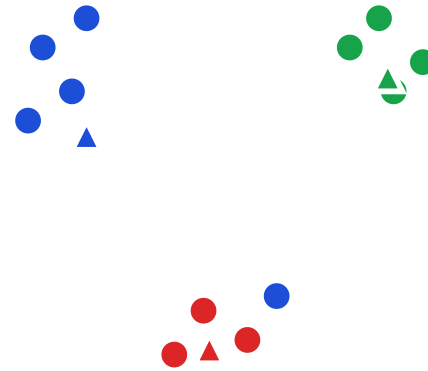
k-Means in action

Same 12 points, three snapshots: random init → mid-iteration → converged.

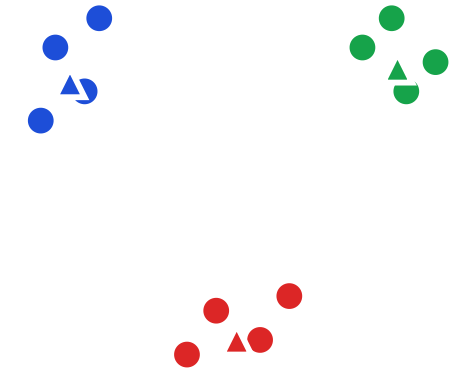
Iteration 0 (random init)



Iteration 2 (mid)



Converged



Centroids drift; assignments stabilize; cost decreases monotonically.

One iteration: a numeric walk-through

$k = 2$, Euclidean (squared) distance. Data + initial centroids:

example	x_1	x_2	x_3
1	0.2	0.5	0
2	-0.6	2.1	1.2
3	-0.5	1.9	1.3
4	0.1	0.5	-0.3

$$c_1 = (0.3, 0.8, -0.5), \quad c_2 = (-0.1, -0.5, 1.0).$$

Assign step (squared distance to each centroid):

$$\text{ex 1: } d^2(c_1) = 0.35, \quad d^2(c_2) = 2.09 \rightarrow$$

c_1

$$\text{ex 2: } d^2(c_1) = 5.39, \quad d^2(c_2) = 7.05 \rightarrow$$

c_1

$$\text{ex 3: } d^2(c_1) = 5.09, \quad d^2(c_2) = 6.01 \rightarrow$$

c_1

$$\text{ex 4: } d^2(c_1) = 0.17, \quad d^2(c_2) = 2.73 \rightarrow$$

c_1

Update step:

$$c'_1 = (-0.2, 1.25, 0.55), \quad c'_2 = \emptyset$$

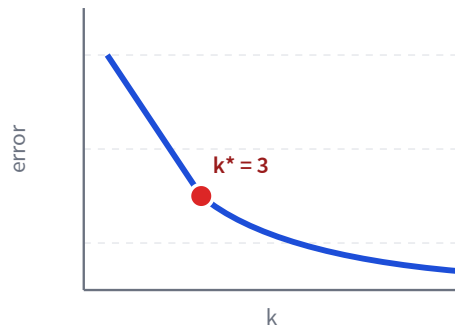
Watch out: c_2 got no examples. Re-initialize it (random restart).

How do we choose k ?

Larger k always gives lower training error — but defeats the point of compression.

Elbow method

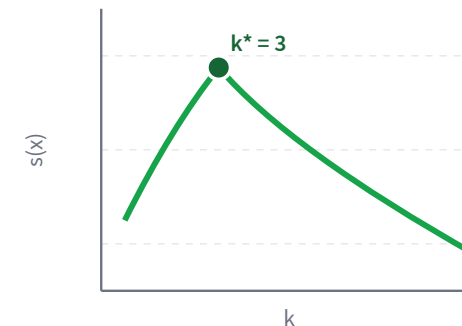
1. Run k-means for $k \in \{1, \dots, k_{\max}\}$.
2. Plot average within-cluster distance.
3. Pick the "elbow" where error stops dropping fast.



Silhouette score

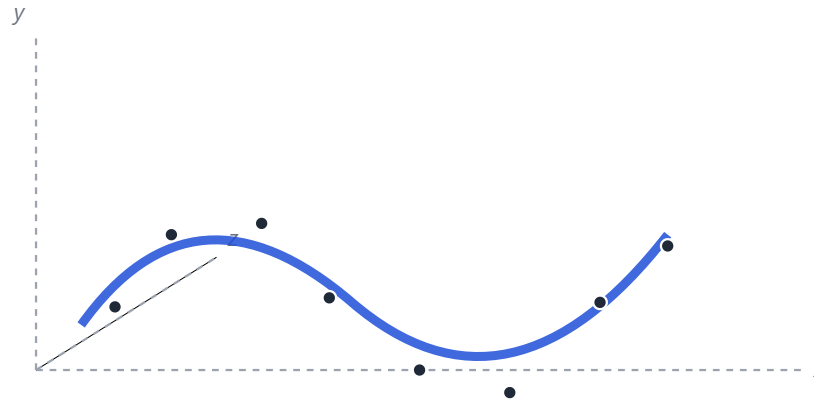
$$s(x) = \frac{b(x) - a(x)}{\max(a(x), b(x))}$$

- $a(x)$ = mean distance to own cluster.
- $b(x)$ = mean distance to *nearest other* cluster.
- Pick k maximizing average s .



Dimension reduction: the manifold idea

High-dimensional data often lives on a **low-dimensional manifold**. Extra dimensions are mostly noise.



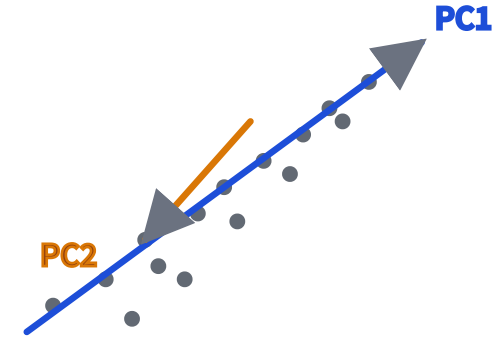
Goal: recover the intrinsic 1D coordinate along the blue ribbon.

Principal Component Analysis (PCA)

The PCA recipe

Find **orthogonal** axes that successively **maximize the variance** of the projected data.

- First PC = direction of largest variance.
- Second PC = direction orthogonal to first that maximizes remaining variance.
- Project onto the top k PCs for a compact representation.



How to compute PCA

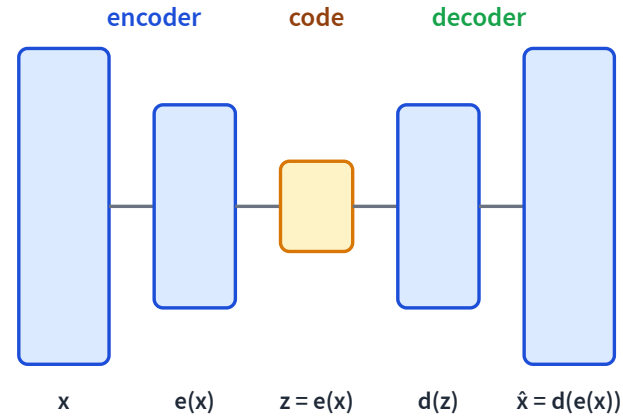
$$\text{PCA}(X, k)$$

1. Mean-center the data: $x \leftarrow x - \text{mean}(x)$.
2. Compute the covariance matrix $\Sigma = \frac{1}{m} X^\top X$.
3. Eigendecompose Σ : $\Sigma v_k = \lambda_k v_k$. Eigenvectors v_k are the principal components, sorted by $\lambda_1 \geq \lambda_2 \geq \dots$
4. Project onto the top k PCs: $z = V_{\text{top-}k}^\top x$.

Fraction of variance kept by the k -th PC: $\frac{\lambda_k}{\sum_j \lambda_j}$. Cumulative across top k PCs guides how many to keep.

Autoencoders: encoder + decoder

A neural net learns to **squeeze and reconstruct** its input.



Loss:

$$E = \sum_i (x_i - d(e(x_i)))^2$$

Encoder + decoder are trained jointly so that the bottleneck z keeps enough info to rebuild x .

Linear vs deep autoencoders

Linear autoencoder

Single linear layer; weights shared between encoder and decoder.

$$\hat{z} = Wx, \quad \hat{x} = W^\top \hat{z}$$

With squared loss, the solution recovers the same subspace as PCA.

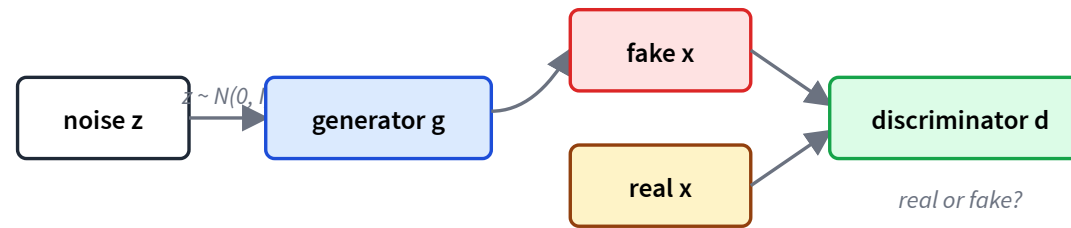
Deep neural autoencoder

Stack non-linear layers (ReLU, etc.) in both encoder and decoder; train end-to-end with backprop.

Captures non-linear manifolds — the modern workhorse for representation learning.

Generative Adversarial Networks (GANs)

Two networks play a game: a **generator** tries to fool a **discriminator**.



- **Generator** $g(z)$: turns Gaussian noise z into a synthetic example that should look real.
- **Discriminator** $d(x)$: binary classifier — was x drawn from training data, or from g ?

GAN training: a minimax game

Both networks optimize the same value, in opposite directions:

$$E = \mathbb{E}_{x \sim p_{\text{data}}} [\log d(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - d(g(z)))]$$

Discriminator

Maximize E : push $d(\text{real}) \rightarrow 1$ and $d(\text{fake}) \rightarrow 0$.

Generator

Minimize E : generate fakes that *fool* the discriminator.

At equilibrium: g produces realistic samples and $d(x) \approx \frac{1}{2}$ — maximally uncertain.

Learning goals (recap) — Next: decision trees

- ✓ Understand what **unsupervised learning** is.
- ✓ Trace the **k-means** clustering algorithm.
- ✓ Perform **PCA** by mean-centering, covariance, and eigendecomposition.
- ✓ Understand the basic idea of **autoencoders** and **GANs**.

L18: back to supervised — the **decision tree** learner.